



Blockchain-Driven Queueing and Transparent Budget Allocation for Scholarships

 **CC 103 - Data Structures and Algorithms**

CS 1206

 JRC Studios

Amelano, Charles V.

Arago, Mary Jobine R.

Bituin, Ralph Ivan I.

College of Informatics and Computing Sciences | Batangas State University | Alangilan Campus

☞★.° Overview

ScholarChain is a console-based C++ application that simulates a **scholarship allocation system** using queueing and blockchain-inspired data structures.

It demonstrates the practical use of **data structures and algorithms**, including stacks, queues, and priority queues, alongside recursion and iteration for efficient processing.

Users can:

-  Apply for scholarships
-  View ranked applicants
-  Process applicants fairly
-  Manage waitlisted entries

Data Handling

-  Applicant records are processed and stored using structured data systems.
-

☞★.° Features

1. **Applicant Evaluation.** Assesses eligibility based on academic and financial criteria
 2. **Priority Ranking.** Uses priority queue for fair distribution
 3. **Queue System.** Processes applicants in FIFO order
 4. **Blockchain Record.** Stores approved applicants chronologically
 5. **Budget Allocation.** Allocates funds until limit is reached
 6. **Waitlist System.** Handles overflow applicants
-

☞★.° How to Run the Program

Open your terminal in the project folder and run:

```
g++ main.cpp queue.cpp priority.cpp stack.cpp hashing.cpp network.cpp -o dsa
```

Run the program using:

```
./dsa
```

🌟 Project Structure

```
DSA/
├── main.cpp
├── README.md
├── .gitignore
├── Data Structure/
│   ├── queue.cpp
│   ├── priority.cpp
│   └── stack.cpp
├── Integrity/
│   ├── hashing.cpp
│   └── network.cpp
└── include
    ├── main.h
    └── datastructs.h
```

- **main.cpp** - Entry point of the program; handles user interaction and system flow
- **.gitignore** - Specifies files to be ignored by Git

📁 Data Structure

- **queue.cpp** - Handles normal applicant processing (FIFO)
- **priority.cpp** - Manages high-priority applicants
- **stack.cpp** - Stores blockchain-like records

📁 Integrity

- **hashing.cpp** - Ensures data integrity using hashing techniques
- **network.cpp** - Simulates blockchain network validation/consensus

📁 include

- **main.h** - contains the applicant structure, applicant information, blockchain-related fields, and automatic priority computation logic

- `datastructs.h` - defines the custom data structures used in the system such as Priority Queue, Dual Queue, and Blockchain Stack

☞★.°Example Output

```
Welcome Admin!

Please set budget before processing!
Budget: 100000

ScholarChain Menu:
1: Process Scholarship
2: Push Queue to Blockchain
3: Show Transactions
4: Show Queue
5: Show Remaining Budget
6: Sort Transaction
7: Verify Transaction Integrity
8: Add Budget
9: Exit Program
Choice (1 - 9): 1

Processing Scholarships
Enter Student ID Number: 25-09705
Enter Student Name: Jobine Arago
Enter Student GPA: 2.50
Enter Student/Family Income: 50000
Enter Scholarship Grant: 2000
Applicant is queued for the normal queue.

Would you like to process another? (Y/N):
```

☞★.°Contributors

Name	Role
Charles V. Amelano	Main Programmer
Mary Jobine R. Arago	Documenter
Ralph Ivan I. Bituin	System Architect

☞★.°Acknowledgement

The team would like to thank Ms. Agdon for her guidance throughout the course until the accomplishment of ScholarChain. Her encouragement and valuable feedback greatly contributed to the development and completion of this system. We also extend our gratitude to our classmates, families, and everyone who supported and inspired us throughout the making of this project.

=====